

Mohamed Elsayed

Software Engineer

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📍 Cairo – Egypt

🇪🇬 Military Service: (COMPLETED)

SUMMARY

Software Engineer with experience building production Django and Spring Boot backends. Expert at System architecture, REST API design, and asynchronous systems (Celery, RabbitMQ). ITI graduate and Ain Shams University alumnus with a strong foundation in data structures, design patterns, and clean architecture principles.

EXPERIENCE

Software Engineer (Full-time)

LightIdea IT Consultancy

December 2025 - present, Cairo, Egypt

- Developed backend services using Django with MVC architecture, applying OOP and design patterns.
- Implemented asynchronous processing using Celery and RabbitMQ to handle background tasks efficiently.
- Designed and optimized SQL databases and queries for a government project (Ministry of Foreign Affairs, Saudi Arabia).

Back-end Developer (Part-time)

White Matter

December 2025 - present, Cairo, Egypt

- Designed and built a scalable multi-tenant SaaS backend using schema-per-tenant isolation, Zero Trust middleware with JWT + RBAC, and layered architecture (API → Application → Domain → Infrastructure) with dependency injection.
- Implemented event-driven communication between bounded contexts via an event bus and message queue, and integrated external systems (identity providers, object storage) using adapter patterns.

Full Stack Using Python

Information Technology Institute (ITI)

July 2025 - November 2025, Cairo, Egypt

- Developed multiple frontend projects using HTML, CSS, and JavaScript, and advanced to building backend applications using Python, Django, and RESTful APIs.
- Completed intensive training covering Linux, Databases (SQL), Problem Solving, Operating Systems, Networks, and Software Testing.
- Collaborated in team-based projects following software development best practices, version control (Git/GitHub), and agile methodologies.

Software Engineer Intern

Orange Innovation Egypt (OIE)

August 2023 - November 2023, Cairo, Egypt

- Learned and applied Spring Boot principles, REST APIs, Build tools (Maven, Gradle), and architectural design patterns (MVC) in real-world projects.
- Developed SQL skills using PostgreSQL, ORMs, and implemented design patterns in Java for side projects.
- Gained hands-on experience working on real-world projects, applying core Java concepts and design patterns.

SKILLS

Programming Languages: Java, Python, JavaScript, C++, SQL

Programming Concepts: OOP, Data Structures, Algorithms, Design Patterns, Problem Solving, OS, Computer Networks

Back-end Technologies: Spring Boot, Django, Spring Security, JWT, REST APIs, Spring MVC, Hibernate, Microservices

Front-end Technologies: React, HTML, CSS, Bootstrap, JavaScript

Database Concepts: ERD, Normalization, Relational

Databases: PostgreSQL, MySQL, SQLite

Testing: JUnit, TestNG, Selenium, Software Testing Fundamentals

Developer Tools: Git, GitHub, Postman, Linux OS

Build Tools: Maven, Gradle

PROJECT

MedFlow Pro — Healthcare SaaS Platform

• June 2026

- Built a full-stack multi-tenant EMR platform covering patient records, appointments, lab orders, billing, pharmacy, and insurance — 15 Django apps with schema-per-tenant data isolation (PostgreSQL).
- Implemented HIPAA-ready security: granular RBAC, Fernet encryption for PII, immutable audit trail, signed (immutable) clinical records, and JWT authentication with Keycloak OIDC support.
- Integrated Celery + RabbitMQ for async AI processing (prescription OCR, lab result analysis), built a React 19 frontend with TanStack Query, Zustand, and Framer Motion, and configured MinIO for S3-compatible document storage.

Online Book Store

• April 2024 - June 2024

- Built an administration and user system using Java and Spring Boot, with features like item, order management, and book/category modification for admins.
- Implemented user functionalities such as sign-in, cart management, reviews, order tracking, filtering, and payment processing.
- Applied design patterns (Builder, Strategy, Factory, Singleton) to enhance the system's architecture and maintainability.

CV Connect

• April 2024 - May 2025

- Developed using Java and Spring Boot, providing functionality for admins to upload CV templates and users to register and log in.
- Enabled job seekers to download CV templates, upload their CVs, and edit their profile and contact information.
- Built HR features to search profiles by skills/job title, filter by education/experience, and review both job seekers' profiles and CVs, streamlining the hiring process.

Use Case Diagram Generator [Graduation Project, Grade: A+]

• April 2024 - June 2024

- Developed an innovative AI-powered software tool using Python, Node.js, HTML, CSS, and JavaScript to automate the creation of use case diagrams from user stories.
- Integrated a T5 base transformer model to process user inputs, extract components of the use case diagram, and generate diagrams instantly.
- Implemented a back-end system to visualize the use case diagrams on a web application, enhancing user experience and efficiency.

EDUCATION

Bachelor of Computer and information Science

Minor in Information System • Ain Shams University • Cairo • 2024

INVOLVEMENT

Science and Technology Committee Member

Ain Shams University • Open-Source Community • January 2023 - July 2024

- Organizing workshops, mentoring teams, delivering sessions, and supporting community growth.
- Gained experience in teamwork, event coordination, and communication skills.

Languages

Arabic & English